

SPORT-SPECIFIC DECISION-MAKING IN A GO/NOGO
REACTION TASK: DIFFERENCE AMONG NONATHLETES
AND BASEBALL AND BASKETBALL PLAYERS¹

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Summary.—The present study examined whether Go/Nogo reaction time (RT) is a relevant index of the sport expertise relating to sport-specific decision-making. 57 male university students, 20 basketball players, 24 baseball players, and 13 sedentary students as a control group, performed a Simple RT task and Go/NoGo RT task which had baseball specific stimulus-response relations. Participants in baseball and basketball differed further in having high, medium, and low experience in the sports. For comparisons across sports, the basketball and the baseball players had significantly shorter reaction times than the nonathletes in both tasks. In contrast, reaction times varied significantly across experience for the baseball players in the Go/NoGo RT task but not for basketball players. These results suggested that Go/NoGo RT could be used as an index of expertise for sport-specific decision-making, if stimulus-response relation in Go/NoGo RT task has a natural relation for a particular sport-domain.

In many sport situations with strict time constraints, athletes are required to judge what is occurring within their surrounding environment as rapidly and accurately as possible, to select an appropriate movement response, and then to organize the selected movement. Therefore, athletic performance is affected by the athletes' information processing from stimulus detection through motor execution (Abernethy, 1996; Schmidt & Wrisberg, 2004). Because reaction time (RT) represents a temporal aspect of the information-processing efficiency (Donders, 1868/1969; Massaro, 1989), it has been used as an index of sport expertise (e.g., Olsen, 1956; Williams & Walmsley, 2000; Wang, Chen, Limroongreungrat, & Change, 2005).

In contrast to the arguments presented in the earlier studies, several researchers have found no relationship between RT and sport expertise (e.g., Abernethy & Neal, 1999; Helsen & Starks, 1999). Abernethy (1996) pointed out that the experts' advantage of information processing is due to perceptual and decision-making skills that are achieved by domain-specific knowledge. Since the RT tasks universally use generic stimuli (e.g., flash of lights) and do not have domain-specific information, RT cannot definitely represent a difference in amount of skill in sport-specific information processing (Aber-

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nethy & Neal, 1999; Mori, Ohtani, & Imanaka, 2002). Thus in research on sport expertise, given the lack of consistency and domain specificity in RT tasks using generic stimuli, the use of RT as an index of sport expertise has been criticized (e.g., Starkes & Deakin, 1984; Abernethy & Neal, 1999).

However, Di Russo, Taddei, Aprile, and Spinelli (2006) examined fencing expertise by using the Go/NoGo RT task of making a decision regarding response execution or inhibition and reported that fencers who are required to make such decisions in response to their opponents' actions, including feints, had shorter RTs to Go stimuli (Go RT) than nonathletes (see also Rossi, Zani, Taddei, & Pesce, 1992). These results may reflect similar decision-making processes in the Go/NoGo task and fencing situation. It is suggested that Go RTs may represent the efficiency of those decision-making processes common to sports requiring decision-making with regard to response execution and inhibition.

To examine whether sports practice improves some sport-specific decision-making abilities, Kida, Oda, and Matsumura (2005) used a light stimulus and conducted the Go/NoGo RT task for baseball players who are required to decide whether to swing as quickly as possible while batting. Baseball players had shorter Go RTs than those of nonathletes and tennis players, and a further difference related to skill was found among the baseball players. Thus, this research showed that despite Go/NoGo RT tasks using generic stimuli with no domain-specific information, a difference was found between athletes (baseball) and nonathletes and also between athletes from two different sports (i.e., tennis and baseball), that is, domain specificity was observed. Their experimental task involved not only deciding responses, which could be considered similar to the execution or inhibition decisions required in baseball at bat but also baseball-specific stimulus-response relationships between the stimulus location and response execution or inhibition. Specifically, if the stimulus was presented to the inside (strike), the reaction was Go (swing), while if it was presented to the outside (out of the strike zone), the reaction was NoGo (not to swing). The phenomenon of finding reduced RT when there is a natural relation between stimulus and response is known as the stimulus-response compatibility effect (Fitts & Seeger, 1953). The above results suggest that the distinct athletic and skill-related differences observed in the baseball domain (Kida, *et al*, 2005) reflect such compatibility.

Therefore, the present study assessed whether a Go/NoGo RT task using generic stimuli can identify sport-specific decision-making performances, based on whether the task exhibited sport-specific stimulus-response relations and on the task characteristic that requires response execution or inhibition. Specifically, a Go/NoGo RT task had a natural stimulus-response relation that was specific to one sport (baseball) and not to the comparison

sport (basketball). In addition, the Go RTs were compared between basketball players, who are required to react to or inhibit the opponents' feints, and baseball players, who not only require response decisions but also the natural stimulus-response relation.

Hypotheses were, first, athletes from the two sports would not differ on simple RTs; however, both groups of athletes would have lower RTs than nonathletes. Second, baseball players would have faster RTs for the Go task than basketball players. Third, baseball players would show significant decreases in the Go RTs according to their skill in the sport; however, this would not hold for basketball players.

METHOD

Participants

A total of 57 male university students, basketball players ($n=20$), baseball players ($n=24$), and sedentary students ($n=13$) as a control group, participated. They all reported normal or corrected-to-normal vision. They were informed of the experimental procedures in advance and consented to take part in this experiment. Players of each sport also were divided into high, medium, and low skill.

Apparatus and Stimuli

All visual stimuli were presented on a color computer display, with a screen size of 40×45 cm. RTs and intertrial interval were controlled with a custom-made computer with attached PCI timerboard. Responses to the visual stimuli were made by key-pressing on a standard keyboard. A jaw restraint was used to hold the subject's eyes level and at a known distance to the display. Visual stimuli were presented on a display located 50 cm in front of each subject's eyes. Four square frames (3×3 cm) of white outlines were presented in a horizontal row on the screen. Their horizontal visual range was 27.4° . Each square and background were black. The visual stimuli were presented by changing at random one of the four squares' color from black to green for 50 msec.

Task and Procedure

The tasks were (a) a Simple RT task on which the participant reacted to the stimulus onset as fast as possible, regardless of the stimulus location by pressing a space key with the dominant index finger; (b) a Go/NoGo RT task on which the participant was asked to respond as quickly and accurately as possible with the dominant index finger when either of the two inside frames were onset (Go signal), but they were required to inhibit the response to the other outside frames (NoGo stimuli).

The participant sat 50 cm away from the display in a dimly lit room and then put his jaw on a stand. There were 40 trials for each task. Before

each task, 20 practice trials were performed to familiarize the subject. The subject took a 5-min. rest between tasks. These tasks were counterbalanced, and the orders of stimulus location and intertrial interval (3, 4, or 5 sec.) were randomized. In any task condition, the probability of stimulus appearance at each location was 25%. RT was the time from stimulus onset to key press and was measured as Simple RT and Go RT in the simple and the Go/NoGo RT task, respectively. Moreover, when the participant responded to NoGo stimuli, the trial was evaluated as an error of commission.

RESULTS

Because there were no significant differences for errors of commission, only RTs were used for subsequent analysis. RTs for all experiments were trimmed so all latencies less or greater than the mean ± 2 SD were eliminated, as were incorrect responses. This resulted in 0.8% of Simple RT and 1.3% of Go RT being trimmed.

Sports Experiences

Table 1 shows the mean Simple RT and Go RT of the baseball players, basketball players, and nonathletes. The mean RTs were subjected to a repeated measures 3 (Groups: baseball, basketball, and control) $\times$ 2 (Tasks: Simple RT task and Go/NoGo RT task) analysis of variance, with repeated measures on task conditions. The two-way analysis of variance showed significant main effect for the Group ($F_{2,54} = 13.42$, $p < .001$, $\eta_p^2 = .33$), Task ($F_{1,54} = 499.01$, $p < .001$, $\eta_p^2 = .90$), and their interaction ($F_{2,54} = 10.11$, $p < .001$, $\eta_p^2 = .27$). An analysis of the simple main effects indicated that the basketball players' responses were significantly shorter than those of nonathletes on Simple RT ($p < .01$, $d = 1.1$) and Go RT ($p < .01$, $d = 1.3$). Also, baseball players showed shorter Simple RT ($p < .01$, $d = 1.1$) and Go RT ($p < .01$, $d = 1.6$) than those of the nonathletes; however, there were no differences between basketball and baseball groups.

TABLE 1
MEAN REACTION TIMES AND STANDARD DEVIATIONS ACROSS GROUPS, SUBGROUPS, AND CONDITIONS

Group		Simple RT		Go RT	
		<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
Basketball	High	232	11	283	6
	Medium	232	13	282	18
	Low	229	22	281	28
	Total	231	15	282	19
Baseball	High	231	12	261	18
	Medium	228	8	272	11
	Low	244	15	296	23
	Total	233	13	274	21
Nonathlete		249	17	317	32

Amount of Skill

To examine the influence of skill-related difference, each group divided into three subgroups. The high- and medium-skilled groups consisted of regular and substitute players, respectively. Participants in the low-skilled group rarely take part in the game [basketball: high ($n=6$), medium ($n=8$), low ($n=6$); baseball: high ($n=9$), medium ($n=9$), low ($n=6$)]. Table 1 shows the mean Simple RT and Go RT of skill-related difference by group. The mean RTs were subjected to a repeated-measures 3 (Groups: baseball, basketball, and control) $\times$ 2 (Tasks: Simple RT task and Go/NoGo Rt task) $\times$ 3 (Skill: high, medium, low) analysis of variance, with repeated measures on task conditions. However, a three-way analysis of variance showed no second-order interaction ($F_{2,38} = 1.60$, $p > .05$, $\eta_p^2 = .08$). Therefore, the mean RTs for each sport separately were analyzed as in previous research (Kida, *et al.*, 2005). For baseball players, a two-way analysis of variance showed significant main effects for Group ($F_{2,21} = 6.93$, $p < .01$, $\eta_p^2 = .40$), Task ($F_{1,21} = 189.36$, $p < .001$, $\eta_p^2 = .90$), and their interaction ($F_{2,21} = 4.30$, $p < .05$, $\eta_p^2 = .29$). An analysis of the simple main effects indicated that the higher-skilled groups had significantly shorter RTs than those of low skill on the Go RT task ($p < .01$, $d = 1.7$) but not on the Simple RT task. This difference also appeared between the medium- and low-skilled groups on Go RT ($p < .01$, $d = 1.3$). In contrast, for basketball, although a two-way analysis of variance showed a significant main effect for Task ($F_{1,21} = 210.93$, $p < .001$, $\eta_p^2 = .93$), there were no significant effects for the group on their interaction.

DISCUSSION

The present study assessed whether a Go/NoGo RT task can identify sport-specific decision-making performance if the task exhibits sport-specific stimulus-response relation in addition to the task characteristic which requires response execution or inhibition.

In the comparison of Simple RT based on sport experience, both baseball and basketball players had superior RTs to the group of nonathletes. Results were consistent with previous findings that Simple RTs of baseball and basketball players were shorter (Olsen, 1956). However, in contrast to a previous study in which only baseball players had faster Go RTs (Kida, *et al.*, 2005), both the baseball and basketball players in this study had shorter Go RTs than those of the group of nonathletes. These results suggest that shorter Go RTs could be observed in sports that include common response execution or inhibition decisions with regard to Go/NoGo RT tasks using light stimuli. However, in the present skill-related differences on Simple RT and Go RT, although baseball players across differently skilled groups had significantly different Go RTs, no such difference was observed among the basketball players with regard to either the Simple RT or Go RT tasks.

A Simple RT is assumed to represent the basic sensory processing efficiency for processes such as stimulus detection and motor execution (Donders, 1868/1969; Massaro, 1989). In addition to this basic sensory processing, Go RT includes both stimulus identification and response selection, which require more cognitive function (e.g., Donders, 1868/1969; Massaro, 1989). In addition, Kida, *et al.* (2005) reported that there is a positive correlation between the Simple RT and Go RT. Thus, the latter is expected to decrease as the former decreases. Based on these notions, the shorter Go and Simple RTs of the baseball and basketball players compared to those of the nonathletes can be interpreted by an inference that athletes show faster basic sensory processing. Moreover, there are skill-related differences in the Go RTs of baseball players, and such differences were not observed with regard to Simple RT. This indicates that highly skilled baseball players not only have faster basic sensory processing but also faster cognitive processing. Such superiority of cognitive processing is consistent with the expectation that a natural relation between stimulus and response should enhance the processing efficiency of perception and response selection (e.g., Zhang & Kornblum, 1998; Proctor & Vu, 2006).

The stimulus-response compatibility effect occurs when there is a physical or conceptual similarity between the stimulus and response sets (Kornblum, Hasbroucq, & Osman, 1990; Kornblum, 1992; Shiu & Kornblum, 1999). That is, baseball players have conceptually linked the domain-specific spatial information and response execution or inhibition, but no such links have been found with respect to the other groups. Therefore, baseball players would have more enhanced responses on the present task, which has the same stimulus-response relations as those in baseball batting. In addition, with respect to the assessment of this facilitation of response by stimulus-response compatibility, the most widely accepted models are dual process models which distinguish between two response-selection routes (Kornblum, *et al.*, 1990; Kornblum & Lee, 1995). According to this perspective, when all the S-R mappings are compatible, response selection occurs via a direct or an automatic route that leads to activation through long-term S-R associations. When the mapping are incompatible, the direct route is suppressed, and the response selection occurs via an intentional translation route. Thus, it appears that these skilled baseball players could automatically select the Go response to the centered stimuli. That is, the skill-related differences among baseball players would reflect different response-selection routes. As another possibility, the skill of baseball players may vary more widely than for basketball players given the difference in the number of players who play regularly in the respective teams. This factor may affect the sport-specific effect.

Further, there was no athletic or skill-related difference in commission

error in response to the NoGo stimuli. Fast responders on Go/NoGo RT tasks must have increased strength of inhibition under NoGo conditions to prevent errors of commission because a fast response on the Go/NoGo RT task typically leads to many errors of commission (Smith, Johnstone, & Barry, 2006). In fact, previous research has suggested that strong response inhibition enables fencers to respond rapidly on Go/NoGo RT tasks (Di Russo, *et al.*, 2006). Thus, it is thought that the shorter Go RTs of basketball and baseball players in the present study might be effected by strong inhibition under NoGo conditions.

In conclusion, it appears experts' advantage in domain-specific decision-making processes can be represented more clearly by including not only response execution and inhibition decision in sports situations but also a sport-specific stimulus-response relationship to a Go/NoGo RT task with generic stimuli. However, the present study did not assess whether basketball players facilitate the Go/NoGo response on a basketball-specific task. Further research would be required to include a task which mimics decisions in basketball and show whether there is an interaction between basketball/baseball players and basketball/baseball tasks.

REFERENCES

- ABERNETHY, B. (1996) Training the visual-perceptual skills of athletes: insights from the study of motor expertise. *The American Journal of Sports Medicine*, 24, S89-S92.
- ABERNETHY, B., & NEAL, R. J. (1999) Visual characteristics of clay target shooters. *Journal of Science and Medicine in Sport*, 2, 1-19.
- DI RUSSO, F., TADDEI, F., APNILE, T., & SPINELLI, D. (2006) Neural correlates of fast stimulus discrimination and response selection in top-level fencers. *Neuroscience Letters*, 408, 113-118.
- DONDERS, F. C. (1868/1969) Over de Snelheid van psychische Processen [On the speed of mental processes]. (W. Koster, Transl.) In W. G. Koster (Ed.), *Attention and performance*. Amsterdam: North Holland. Pp. 412-431.
- FITTS, P. M., & SEEGER, C. M. (1953) S-R compatibility: spatial characteristics of stimulus and response codes. *Journal of Experimental Psychology*, 46, 199-210.
- HELSEN, W. F., & STARKES, J. L. (1999) A multidimensional approach to skilled perception and performance in sport. *Applied Cognitive Psychology*, 13, 1-27.
- KIDA, N., ODA, S., & MATSUMURA, M. (2005) Intensive baseball practice improves the Go/Nogo reaction time, but not the simple reaction time. *Cognitive Brain Research*, 22, 257-264.
- KORNBLUM, S. (1992) Dimensional overlap and dimensional relevance in stimulus-response and stimulus-stimulus compatibility. In G. E. Stelmach & J. Requin (Eds.), *Tutorials in motor behavior II*. Amsterdam: Elsevier. Pp. 743-777.
- KORNBLUM, S., HASBROUCQ, T., & OSMAN, A. (1990) Dimensional overlap: cognitive basis for stimulus-response compatibility: a model and taxonomy. *Psychological Review*, 97, 253-270.
- KORNBLUM, S., & LEE, J. W. (1995) Stimulus-response compatibility with relevant and irrelevant stimulus dimensions that do and do not overlap with the response. *Journal of Experimental Psychology: Human Perception and Performance*, 21, 855-875.
- MASSARO, D. W. (1989) *Experimental psychology: an information processing approach*. San Diego, CA: Harcourt Brace Jovanovich.
- MORI, S., OHTANI, Y., & IMANAKA, K. (2002) Reaction times and anticipatory skills of karate athletes. *Human Movement Science*, 21, 213-230.

- OLSEN, E. A., (1956) Relationship between psychological capacities and success in college athletics. *Research Quarterly*, 27, 79-89.
- PROCTOR, R. W., & VU, K. P. L. (2006) *Stimulus-response compatibility principles: data, theory, and application*. Boca Raton, FL: CRC/Taylor & Francis.
- ROSSI, B., ZANI, A., TADDEI, R., & PESCE, C. (1992) Chronometric aspects of information processing in high level fencers as compared to non-athletes: an ERPs and RT study. *Journal of Human Movement Studies*, 23, 17-28.
- SCHMIDT, R. A., & WRISBERG, C. A. (2004) *Motor learning and performance: a problem-based learning approach*. (3rd ed.) Champaign, IL: Human Kinetics.
- SHIU, L., & KORNBLUM, S. (1999) Stimulus-response compatibility effects in go-no-go tasks: a dimensional overlap account. *Perception & Psychophysics*, 61, 1613-1623.
- SMITH, J. L., JOHNSTONE, S. J., & BARRY, R. J. (2006) Effects of pre-stimulus processing on subsequent events in a warned Go/NoGo paradigm: response preparation, execution and inhibition. *International Journal of Psychophysiology*, 61, 121-133.
- STARKES, J. L., & DEAKIN, J. M. (1984) Perception in sport: a cognitive approach to skilled performance. In W. F. Straub & J. M. Williams (Eds.), *Cognitive sport psychology*. Lansing, MI: Sport Science Associates. Pp. 115-128.
- WANG, Y. T., CHEN, S., LIMROONGREUNGRAT, W., & CHANGE, L. (2005) Contributions of selected fundamental factors to wheelchair basketball performance. *Medicine & Science in Sports & Exercise*, 37, 130-137.
- WILLIAMS, L. R. T., & WALMSLEY, A. (2000) Response timing and muscular coordination in fencing: a comparison of elite and novice fencers. *Journal of Science and Medicine in Sport*, 3, 460-475.
- ZHANG, H., & KORNBLUM, S. (1998) The effects of stimulus-response mapping and irrelevant stimulus-response and stimulus-stimulus overlap in four-choice Stroop tasks with single-carrier stimuli. *Journal of Experimental Psychology: Human Perception and Performance*, 24, 3-19.

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